



Wall climbing pads

A team at Stanford University has built a pair of Gecko-inspired wall climbing pads, where testing is still underway <http://dgit.in/WallClimb>

Blood clot injections

Researchers at MIT and Texas A&M University have created a biodegradable gelatin injection that would cut down on blood loss <http://dgit.in/HydrogelInj>

living on the other side of the planet. That being said, advancements in mainstream electronics will continue to grab all the important headlines: How good is the new iPhone? Is the latest version of Android worth the upgrade? Why are my new Xbox games so terrible and more bug-ridden than my latest build of Linux?

Amidst this mess of consumer gadgets and internet services, most of us pay little heed to the ground-breaking advances made in the field of robotics, some of which would undoubtedly leave you having to pick up your jaw off the floor. Although there are a myriad of examples to exhibit, here are some that in our opinion have stood out in terms of scale as well as sheer brilliance and innovation.

Mechanised mules

Fewer sights this year have been as technologically awe-inspiring and eerily nerve-jangling than the sight of Boston Dynamic's ground-breaking robotics project, BigDog. To the uninitiated, this intimidating robot looks better suited for the sets of a horror movie. However, the engine-powered hydraulic actuation system that controls its four legs make it clear that this 150-kilogram metallic marvel is a testament to how close we are to perfecting robots that can accurately replicate the bionic movements of animals.

BigDog is no small fry to operate, though. It's packed with close to 50 sensors, many of which are used to measure a massive array of parameters – ranging from attitude, acceleration, engine speed, and temperature to the motion and force of joint actuators and the internal engine's

hydraulic pressure. Controls are conveyed through the on-board internal computer, which processes signals that vary the position and force of the joints as well as their velocity and altitude during locomotion.

BigDog is funded by the US-based Defence Advanced Research Projects Agency (DARPA) with an aim to serve as a pack mule that can accompany soldiers on rough terrains that conventional ATVs cannot traverse. The project has seen tremendous improvements over the course of its development. In fact, in 2013, Boston Dynamics released video footage of a modified BigDog that possessed an arm with the ability to pick up and throw large objects weighing up to 23 kilograms.

This project is just one of many attempts of the US army to replace grunts with robots. Some more examples are the well known unmanned drones and the lesser known sentry tech guns, such as the Phalanx CIWS – an automated weapons system for defence against anti-ship missiles. If there ever was an organisation that had a love affair with Robotics, it's the (specifically American) military.

Robo-raptors

East Asian countries such as South Korea and Japan have always been at the forefront of cutting-edge developments in the field of robotics. South Korea, in particular, has served up some extremely interesting bionic oddities – from humanoid robots that replace fans at stadiums by duplicating the actions and emotions of their remote human counterparts at home to

deadly machine gun-toting and grenade-launching robot sentries such as the SGR-A1, which detects intruders using cameras and heat and motion sensors.

Similarly, researchers and engineers from the MSC Lab at KAIST – a public research university in South Korea – have engineered a robot that's best described as a fast-running biped robot capable of reaching speeds of up to 46 kilometres per hour on a treadmill with off-board power. Dubbed as the 'Raptor', it

has a functional design heavily inspired by the velociraptor (insert Jurassic Park reference here), with two under-actuated legs and a tail, which is used for assisted pitch control that provides stability over high obstacles.

Many will highlight the similarities between the Raptor and the world's current faster robot, Boston Dynamic's



Boston Dynamic's WildCat looking for the next dog it can send scampering

very own 'Cheetah', which has managed to record an astounding 49 kilometres per hour. Despite their similar speeds, however, the two robots are strikingly different. The Cheetah is a hefty quadruped powered by hydraulic actuators, while the Raptor is a compact biped 3-kilogram automaton with legs made of a lightweight composite material.

However, these robots have a long way to go in comparison to land-tested robots such as BigDog, considering they've only been tested on treadmills while supported by a mechanical rod for safety. Boston Dynamics hopes to test and unveil a new version of the Cheetah known as WildCat, which should be able to run independently outdoors by the start of 2015.

Animatronic arachnids

When we think of animatronic robots, we generally think in the scale of bots the size of domestic dogs; because who among us hasn't fantasized about owning our very own pet robot? Unsurprisingly, very few think of 'pets' along the scale of more manoeuvrable insect-sized mechanoids.



Poop scraper not included